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Comparison of deconvolution algorithms for GeoMx spatial transcriptomics immune cell data

Introduction

Spatial Transcriptomics (ST) is a powerful NGS-based technique that combines expression information and spatial context, allowing new insights into the cellular composition and spatial organization of tissues. Since spot-based ST data and the marker-based NanoString GeoMx technology are typically limited to multi-cell resolution, a precise estimation of the abundance of cell types per region is essential for accurate ST data interpretation. Several deconvolution algorithms for bulk RNA-seq or ST data have been published, but their performance is depending on the input data, reference profiles, and algorithm settings, and a comprehensive benchmarking of deconvolution algorithms for NanoString GeoMx immune cell data is still lacking.

Methods

We compared the performance of SpatialDecon [1], FARDEEP [2] and EPIC [3] regarding the inference of immune cell types on GeoMx data from CD68-marked multiple sclerosis (MS) lesions and public lung tumor samples [1] (Fig.1). We used two different immune cell profiles for the cell type deconvolution, and assessed effects of data quality and algorithm settings.

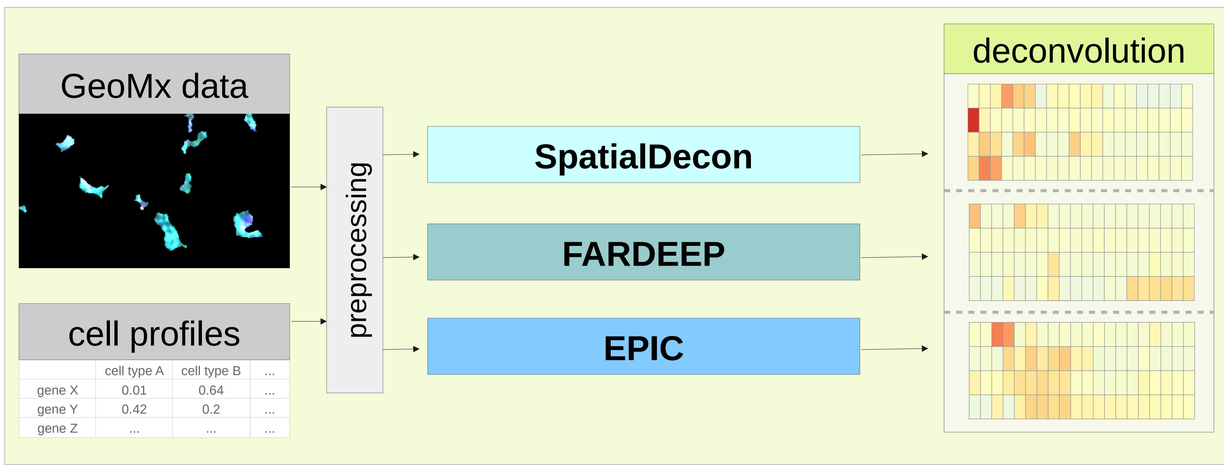

Figure 1: Benchmarkinig workflow. After preprocessing the GeoMx data with GeomxTools, deconvolution results were calculated for the three algorithms SpatialDecon, FARDEEP, and EPIC.

Results

Deconvolution patterns were not correlated to basic quality markers like the number of detected genes for all MS CD68 samples, if a general minimum sample quality was given. Detection of cell populations was depending on the immune cell profiles, with subpopulations being lost if more general signatures were used. For the MS data, FARDEEP and SpatialDecon yielded more similar deconvolution results than EPIC (Fig.2) and detected a higher proportion of macrophages, while EPIC's inclusion of “other cells” allowed a more accurate deconvolution of control samples. For the lung dataset, all programs found major immune cell populations in the TME samples.

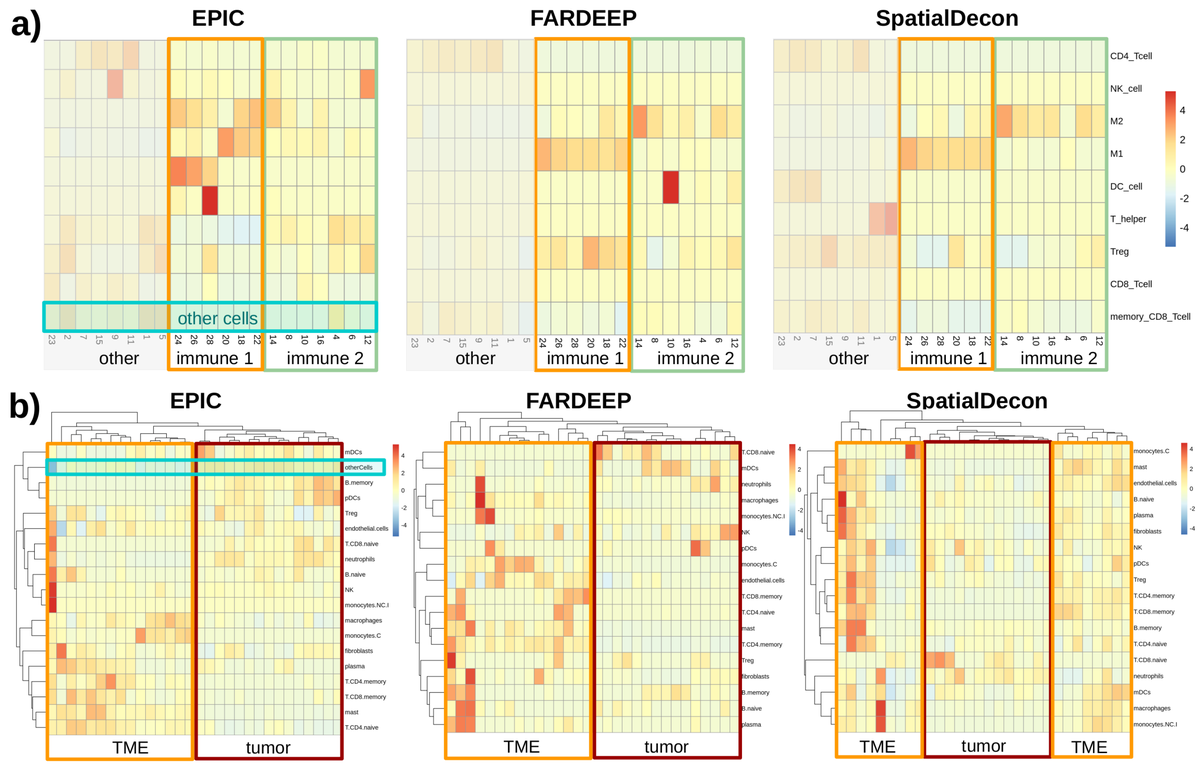

Figure 2: Deconvolution heatmaps, based on a) MS data and immune signatures (ordered per group for easier comparison), b) lung cancer data and immune signatures (default clustering).

Discussion & Conclusion

FARDEEP, SpatialDecon, and EPIC consistently identified core immune cell populations in all chosen data sets. Detection of other cell types was more variable, and depended on the algorithm and reference immune profiles. For mismatches between input data and cell profiles, the flexible tool EPIC offered a more realistic deconvolution through the use of 'other cells'. A more thorough benchmarking study is required to optimize the deconvolution of GeoMx data in different tissues.

Literature

[1] Danaher et al. (2022) Advances in mixed cell deconvolution enable quantification of cell types in spatial transcriptomic data, Nat Commun, 13(1):385. [2] Hao et al. (2019) Fast and robust deconvolution of tumor infiltrating lymphocyte from expression profiles using least trimmed squares, PLoS Comput Biol, 15(5):e1006976. [3] Racle et al. (2020) EPIC: A Tool to Estimate the Proportions of Different Cell Types from Bulk Gene Expression Data, Methods Mol Biol, 2120:233-248.

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